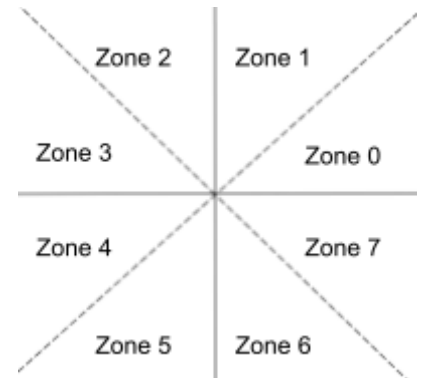


Name: _____ ID: _____ Section: _____

1. What advantages does the Midpoint Line (MPL) Algorithm offer over the Digital Differential Analyzer (DDA) Algorithm in line drawing? 2

2. While using Eight-Way Symmetry, we divide the coordinate space into 8 zones based on the slope and direction of the line.

- a. Determine the zone of the line segment starting from (11, 20) and ending at (2, 12) using Eight-Way Symmetry. Show calculations and justify your answer. 3
- b. If we convert a point (5, 9) from Zone 6 to Zone 0 using Eight-Way Symmetry, what will be the output coordinates? 3



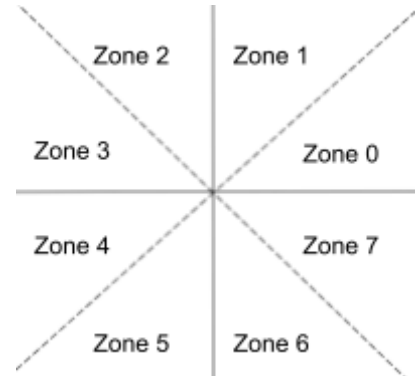
3. Using the Midpoint Line Drawing Algorithm, find the first 6 “missing” pixel values starting from (2, 3) to (19, 7). Show your calculations. 12

Name: _____ ID: _____ Section: _____

1. What advantages does the Midpoint Line (MPL) Algorithm offer over the Digital Differential Analyzer (DDA) Algorithm in line drawing? 2

2. While using Eight-Way Symmetry, we divide the coordinate space into 8 zones based on the slope and direction of the line.

- a. Determine the zone of the line segment starting from (9, 19) and ending at (2, 13) using Eight-Way Symmetry. Show calculations and justify your answer. 3
- b. If we convert a point (5,9) from Zone 2 to Zone 0 using Eight-Way Symmetry, what will be the output coordinates? 3



3. Using the Midpoint Line Drawing Algorithm, find the first 6 “missing” pixel values starting from (1, 4) to (17, 11). Show your calculations. 12